THE DIRECT IMAGE ALONG A SET-VALUED FUNCTOR IS RIGHT ADJOINT TO THE STALK FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The description of the stalk of a presheaf. Same chapter, the description of the stalk as the filtered colimit over the pairs consisting of an object and an element, used around Equation equation-stalk. Supplies the description of the stalk functor whose adjointness is asserted.
- The direct image on presheaves. Same chapter, the definition of the functor on presheaves attached to a set-valued functor. Supplies the functor shown to be right adjoint.

Source reference. Original equation 04EH: sources/stacks-sites/pinned-original-source.tex, source lines 8191–8191. Navigation pointer invoked at proof line 8191; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *For any functor $u : \mathcal{C} \to Sets$. The functor u^p is a right adjoint to the stalk functor on presheaves.*

Proof. Let $\mathcal{F}$ be a presheaf on $\mathcal{C}$. Let E be a set. A morphism $\mathcal{F} \to u^p E$ is given by a compatible system of maps $\mathcal{F}(U) \to \mathrm{Map}(u(U), E)$, i.e., a compatible system of maps $\mathcal{F}(U) \times u(U) \to E$. And by definition of $\mathcal{F}_p$ a map $\mathcal{F}_p \to E$ is given by a rule associating with each triple (U, x, σ) an element in E such that equivalent triples

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map to the same element, see discussion surrounding Equation (04EH). This also means a compatible system of maps $\mathcal{F}(U) \times u(U) \to E$. $\square$